

Fig. 2: An actual-size, single-side PCB for sensitive LPG sensor

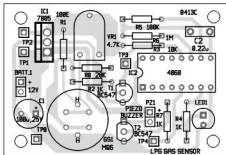


Fig. 3: Component layout for the PCB

the reset pin is high IC2 does not oscillate, and when this pin goes low IC2 starts oscillating.

Working of the circuit is simple. When the sensor detects LPG in the air, its output becomes high and transistor T1 conducts to make reset pin of IC2 low. This triggers IC2 to oscillate, which is indicated by LED1. After a few seconds, the buzzer starts beeping to indicate gas leakage.

The circuit works off 12V DC from a battery (BATT.1) or you can use an adaptor. IC1 provides regulated 5V DC supply for the sensor and IC2.

Construction and testing

An actual-size, single-side PCB for sensitive LPG sensor is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable case with an opening to allow the gas to enter. Place the unit near the LPG cylinder or gas stove within a distance of one metre. Vary preset VR1 to adjust the sensitivity of the sensor.

To test the circuit, check 12V at test point TP1 with respect to TP0 to

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - 4060 ripple-carry binary counter
- T1, T2 - BC547 npn transistor
- LED1 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 100-ohm
- R2, R4, R7 - 1-kilo-ohm
- R3 - 10-kilo-ohm
- R5 - 100-kilo-ohm
- R6 - 1-mega-ohm
- R8 - 20-kilo-ohm
- VR1 - 4.7-kilo-ohm preset

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2 - 0.22 μ F ceramic disk

Miscellaneous:

- BATT.1 - 12V battery
- PZ1 - Piezobuzzer
- GS1 - 6-pin MQ6 LPG sensor

verify the correct power supply. Place the unit near the gas stove burner and turn on the burner for a few seconds without igniting. Then, turn 'the burner off' and adjust VR1 until you see LED1 glowing. TP3 should be low at this moment. ●

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